

CellStream — pilot customer experience

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The announcement in October 1996 of BT's pilot CellStream asynchronous transfer mode (ATM) service marked a significant milestone in the provision of broadband services within the UK telecommunications market. Building on the success of BT's initial broadband Switched Multi-megabit Data Service (SMDS), CellStream provides BT's customers with high-speed and operationally flexible communications within the UK and internationally. This paper outlines BT's CellStream ATM services, and, by taking a particular example, illustrates how customers have been supported in their introduction to them.

1. Introduction

ATM has proved to be a fascinating meeting ground for two major communications groupings — the data networking world (characterised by free market competition and selling equipment to business customers), and the telephone operator world (characterised by standards, scale and selling services).

We are now at the point where customers are beginning to grapple with the opportunities offered by asynchronous transfer mode (ATM) while having to cope with the complexity of a rapidly developing new technology. Operators, such as BT, need to be able to provide a lead to customers by offering sensible services and guidance that can enhance the way they do business.

This paper outlines BT's CellStream ATM services, both national and international, including connection to another operator in the USA. Its main purpose is to illustrate how customers have been supported in their introduction to ATM services, by means of a case study involving NETCOM Internet Limited, CellStream's first international customer.

A basic tutorial on ATM can be found in Adams [1].

2. The CellStream service

2.1 Background

By the mid-1980s it was becoming clear that the existing public networks based on 64 kbit/s channels would be too restrictive for the range of services being envisaged for the future. The emergence of local area network (LAN) technology, video and voice coding techniques, and

multimedia documents would require a wider range of bandwidths to be switched and more flexibility in the allocation of network resources.

Work at BT Laboratories and CNET in France led to the idea of using short, fixed-length packets (or cells) as a common means of supporting all services. In BT, 'Orwell' was developed as a protocol for sharing out bandwidth to meet the differing requirements of voice and data [2]. CNET developed a large ATM cross-point switch called 'Prelude'. Interest in the concept grew rapidly through various RACE projects and ITU contributions until ATM was standardised as the transport mechanism for broadband ISDN.

In the USA it was the data community who latched on to ATM as a means of creating flexible, high-bandwidth LANs. ATM really began to take off world-wide when the ATM Forum was established in 1991 to promote the specification of ATM for local area use. A huge range of ATM products for both local and wide area networks are now available.

In Europe, BT, together with other operators, successfully established a European ATM pilot scheme which supported a wide range of experimental applications. The experience gained from this pilot scheme and work carried out under corporate and other projects provided valuable understanding of how to support voice and data services on ATM networks [1].

Following the European ATM pilot scheme and other significant network trials around the world, network operators are now in the early stages of offering services

based on ATM networks. BT launched its first such service, called CellStream, in March 1997.

2.2 Drivers for CellStream

Over recent years businesses have been taking advantage of high-performance local area communications based on networks of PCs and file servers. There has been enormous growth in the use of electronic documents and e-mail, and more recently the development of the Internet has allowed simple access to multimedia documents. With more and more crucial information being stored electronically, a further demand for remote, secure back-up of data has emerged. All this has led to a need for higher bandwidth and for wide area networks to have performance equivalent to that available locally.

Hence, BT's CellStream range of ATM services was developed in response to a demand from customers for flexible, high-bandwidth connectivity between widely separated sites. In addition, they wanted to have confidence that the service would have a future and be capable of upgrade to accommodate a wide range of applications.

A further driver for BT was the opportunity to begin creating a single multiservice platform to support not only CellStream but also other services such as SMDS, Frame Relay and Internet, with resultant economies of scale.

2.3 Service description

Current service features

Release 1 of BT's CellStream ATM service provides transport of ATM cells over either virtual paths (VPs) or virtual channels (VCs) on a point-to-point virtual connection basis, as shown in Fig 1. It is a sequence-preserving, connection-oriented service with agreed quality of service (QoS) and throughput.

The service is available on a national basis in the UK and, on a correspondent basis, to the USA. It is also planned to connect to a number of other operators' ATM networks within Europe and to provide global ATM via Concert.

The VPs and VCs are available as permanent virtual connections (PVCs). For either case PVCs are managed with agreed QoS and traffic parameters for either constant bit rate (CBR) or variable bit rate (VBR) traffic. Information transfer (i.e. customer traffic) is thus accomplished via a predefined connection across the network. The connection is established by network management operation at subscription time or as part of a reconfiguration activity. The PVC remains established independent of traffic activity and does not require real-time signalling prior to transmission of data across the connection.

Where a virtual path service is provided the customer is free to allocate and manage virtual channels within the virtual path without BT intervention. Traffic management of the VCs, to ensure that overall traffic does not exceed the

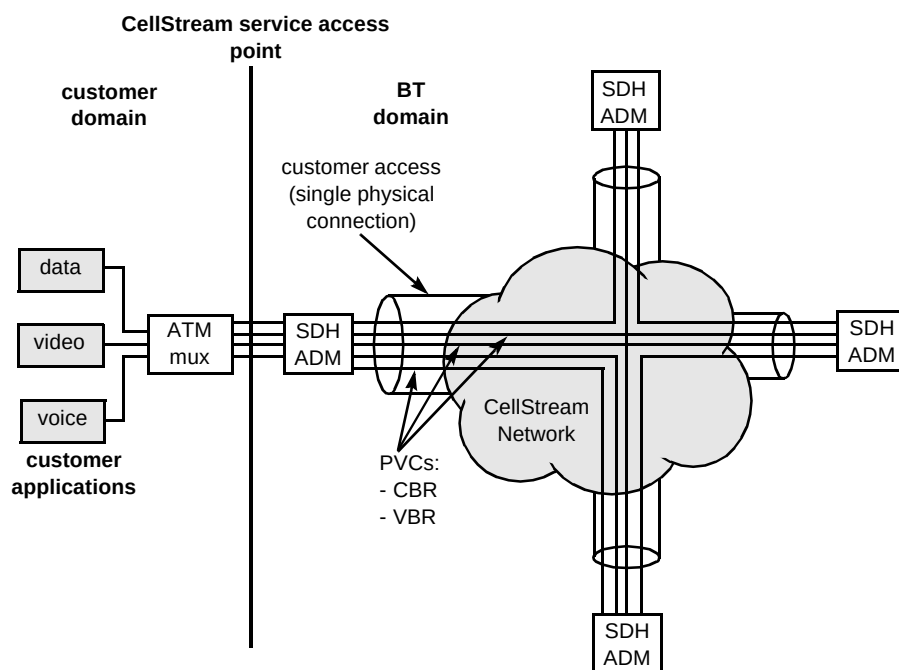


Fig 1 ATM PVCs.

traffic parameters for the virtual path, is the responsibility of the customer. It should be noted that the connection type of all VCs within the VP are the same as that requested for the VP. Traffic volumes are policed at the ingress of the network and any traffic received at a level in excess of agreed parameters is discarded or tagged depending on the service.

Alternatively BT can provide a virtual channel service where the QoS and traffic parameters are managed for individual virtual channels.

BT's CellStream ATM service currently offers the following:

- standards compliance — ATM Forum UNI 3.1,
- customer access options — 2, 34, 155 Mbit/s ATM UNI,
- connection support — supports PVCs (VPs and VCs),
- service support — constant bit rate, variable bit rate,
- traffic descriptors — peak cell rate (maximum) and sustainable cell rate (average): 5 kbit/s to 149 Mbit/s.

Enhanced features

Following the successful launch of the basic CellStream service, a number of enhancements are being considered for future releases. These include:

- options for increased access resilience,
- switched connections,
- increased international connectivity,
- Frame Relay interworking,
- usage-based billing,
- customer network management,
- metropolitan service.

Service release dates

The dates given below are those for the early releases of CellStream. In future it is likely that BT's ATM platform will have two scheduled upgrades per year to support not only new releases of CellStream, but also to support other services as required.

- UK Service, pilot service — October 1996
- USA Service, pilot service — December 1996
- UK Service launch — March 1997
- USA Service launch — June 1997

2.4 Network platform

UK CellStream network

The CellStream PVCs are supported on a high-speed core network based on ATM switching, synchronous digital hierarchy (SDH) transmission technology and optical fibre. SDH and optical fibre are also used to provide the access links to customer premises with bandwidths of 34 Mbit/s and 155 Mbit/s in the UK. Lower speed access at 2 Mbit/s is provided across the plesiochronous digital hierarchy (PDH) network. The network terminating point (NTP) on the customer's premises is the point of interconnect between the network terminating equipment (NTE) and the ATM customer premises equipment (CPE) interface and is the ATM UNI for this service. The ATM service allows connectivity using virtual connections from any NTP to any other NTP on the network.

Figure 2 provides an illustration of part of the network and shows four ATM switches interconnected by SDH with access to a single customer site. Links to Europe and USA are also shown.

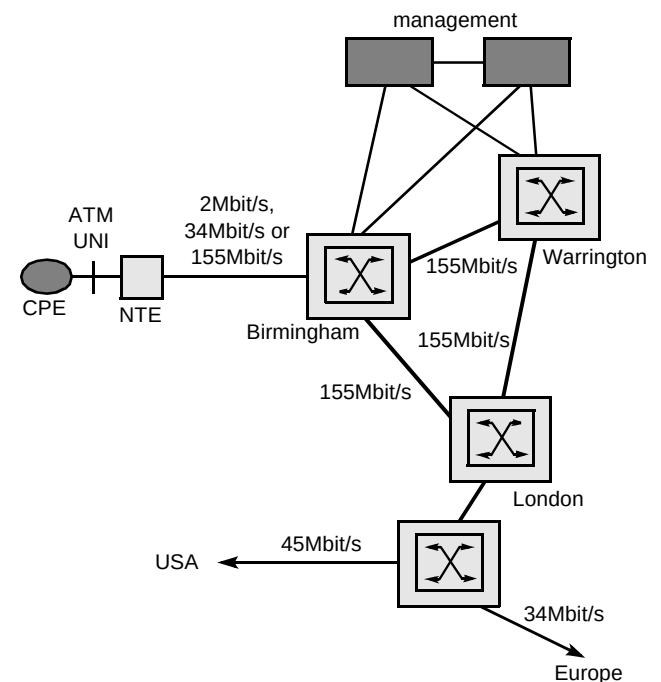


Fig 2 Illustration of part of the CellStream network.

The network is managed from a single network operations unit (NOU) with a stand-by site equipped to take over in the event of a failure. Customer requests and any problems are handled through two advanced service centres with access to BT's operations systems and the NOU.

The network is being expanded as more customers and sites are connected. At the time of writing 16 switches (Newbridge 36170) were deployed providing coverage of the major UK conurbations. The approximate locations are shown in Fig 3. Some locations have more than one switch.

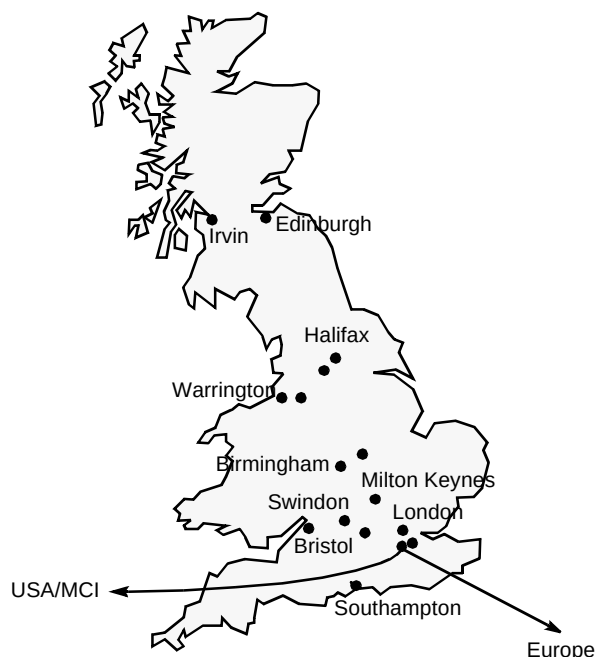


Fig 3 Current CellStream network.

International ATM network

To give CellStream customers the option of international service BT has developed a correspondent-based service to the USA with MCI. The service currently operates with 45 Mbit/s capacity on the transatlantic links known as TAT-12/TAT-13 with ATM gateway switches in London and New Jersey.

International UK ATM service is currently available in the USA from nearly 500 points of presence.

Future plans for the international service include expanding to Canada, other countries in Europe, and the Asia/Pacific regions, while maintaining the same high-quality, robust feature set as is currently available on the domestic services.

3. CellStream customer case study

Previous sections of this paper have covered the background to and drivers for CellStream, and described the BT CellStream national and international ATM services and their underlying network platforms. The CellStream trial commenced in October 1996 and one of the early customers was NETCOM On-line Communication Services Inc and its UK subsidiary NETCOM Internet Ltd. NETCOM are atypical of the majority of customers on the

CellStream network in that they presented BT with a unique business scenario which needed to be satisfied to exacting service requirements; it is for this reason that they are the subject of the following case study.

3.1 NETCOM Internet Ltd

NETCOM On-line Communication Services Inc is a leading international Internet service provider (ISP) in North America. Founded in 1988, it began by providing college students with an off-campus alternative for network access to university computer facilities. As the company grew it was quick to recognise and adapt to the needs of an expanding customer base by providing broader Internet access services. Today NETCOM is a multinational company with subsidiaries based in Toronto, Canada, Sao Paulo, Brazil, and Bracknell, UK.

Business Drivers

The explosive interest and growth in the Internet worldwide over the past 12-18 months has meant that, for companies like NETCOM, keeping pace with capacity requirements, so that customer expectation and service level agreements (SLAs) can be managed, has become very difficult — this is especially problematical when forecasting the needs for international capacity as it is expensive and requires weeks of advanced planning.

In the past when customers have complained to their ISP of inadequate service (which may have manifested itself as slow response times) the cause has been, in part, attributed to network congestion. ISPs have therefore traditionally solved the problem by throwing more bandwidth into their networks, thus forcing them to procure additional capacity for their network — the network being provided through private leased circuits (PLCs).

The downside of using PLCs is that they have to be treated as a single pipe. To provide higher capacity they either have to be multiplexed in parallel or much larger pipes need to be used. Companies are therefore concerned about finding themselves locked into a fixed bandwidth solution which is inadequate and inflexible for future growth.

In NETCOM's case they also had to look to differentiate themselves from their competitors in this very fast moving and competitive sector of the market. This differentiator came in the form of putting quality of service foremost, rather than low-cost access.

Also, having a US parent company, with its own network infrastructure across the US, meant that if NETCOM UK could provide a large enough pipe between itself and the USA over which to run its core Internet protocol (IP) network, it could provide an independent

network between the US and UK thereby giving it total control of traffic with guaranteed bandwidth and service levels — an obvious competitive advantage.

With these factors in mind NETCOM quickly recognised the benefits that could be gained by using asynchronous transfer mode technology and BT's CellStream service between the UK and the USA. To push ahead with its strategy the CellStream service needed to satisfy the following requirements:

- service feature flexibility (bandwidth granularity),
- support for asymmetric permanent virtual connections (PVCs),
- fast reconfiguration (service parameters),
- international service,
- scalability (bandwidth, reach, number of sites),
- network security through resilience,
- cost effectiveness/reduction (compared with PLCs and their international equivalents).

With these requirements in mind and extensive dialogue between NETCOM and BT, the network was designed, implemented, tested and handed over to the customer within a period of six weeks from receipt of order (see Fig 4), proving to NETCOM that, though a new service, BT could deliver the required solution extremely quickly.

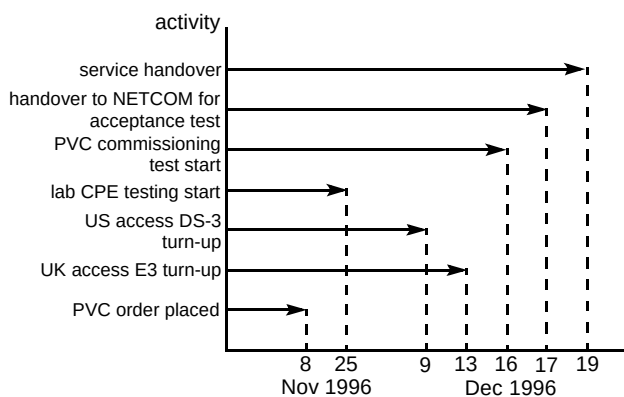


Fig 4 Implementation time-scales.

Customer network

This section describes NETCOM's pre- and post CellStream network, looks at network requirements, describes some of the joint testing undertaken and considers some of the key benefits gained by NETCOM through the use of CellStream.

To support their UK customers NETCOM's pre-CellStream network consisted of high-speed data rings which formed the core backbone. Server and cache machines and various router networks interconnect to the backbone and ultimately to the NETCOM intranet via a firewall. A multi-supplier approach had been taken to provide private leased lines in the UK and internationally (see Fig 5). NETCOM's Bracknell HQ is connected through to Telehouse in London which is the main Internet exchange in the UK. In the US, NETCOM was already a user of an ATM-based service.

With CellStream, the aim was to give NETCOM the ability to interconnect into the US-based ATM network and also the option of replacing their IPLCs, which were costing tens of thousands of pounds a year, with a flexible, reliable and cost-effective service. Figure 6 illustrates the CellStream solution implemented.

An unusual feature of the solution provided to NETCOM was the provision of asymmetric constant bit rate (CBR) ATM PVCs. Bandwidth allocations reflected the nature of the service provided by NETCOM in that twice as much was provisioned in the US-to-UK direction than in the opposite direction; this is logical as the majority of Internet servers are located in the US, hence there is more bandwidth to return files.

Laboratory testing

In the run-up to providing CellStream, BT provided the opportunity for trial customers to carry out joint testing of ATM CPE (in NETCOM's case the Cascade 8000 ATM switch) against CellStream at BT Laboratories (BTL). This allowed both NETCOM and BT to test 'real' applications across the CellStream international network and provided NETCOM with some confidence that their chosen CPE would interoperate with CellStream. The configuration used for this 'validation' is shown in Fig 7. A physical loop in Dallas was provided by MCI to emulate switching in the US.

Testing at BTL of the Cascade 8000 simulated a typical service scenario faced by NETCOM, i.e. client/server. Testing included:

- ping tests between the two PCs (user end-stations),
- file transfers (large files simultaneously in both directions),
- measurement of delay and delay transfer times.

In general, these tests proved that CellStream could efficiently support NETCOM's application services.

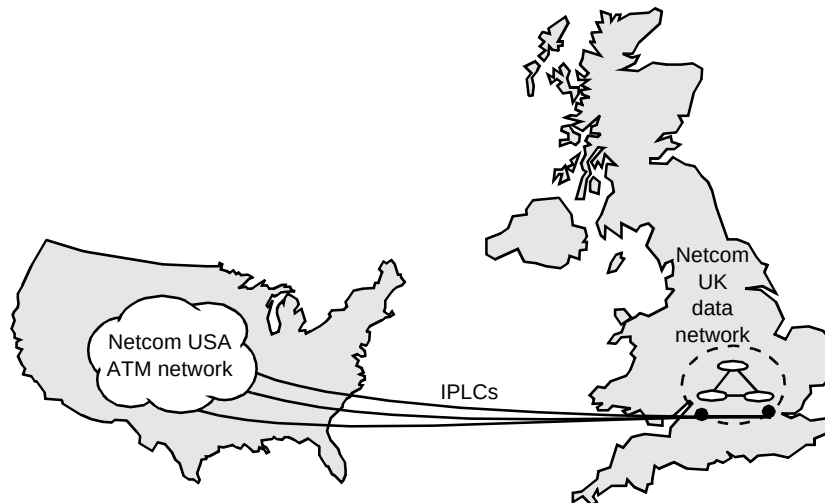


Fig 5 International private leased line solution (IPLC).

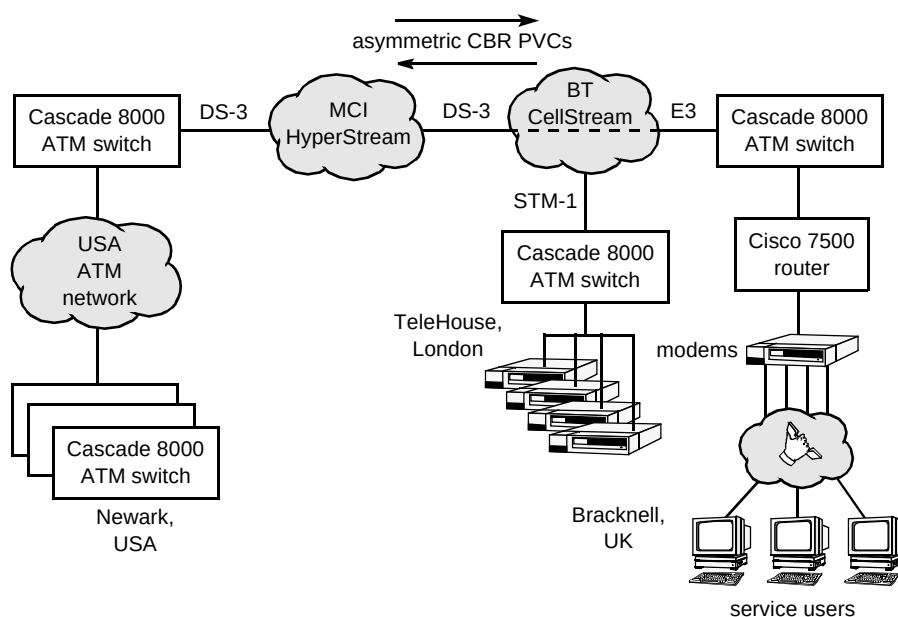


Fig 6 CellStream solution.

Interestingly, a number of issues were revealed which users need to be aware of when implementing ATM, in particular:

- correct setting of traffic policing (PCR, cell delay variation (CDV)),
- traffic shaping configuration and implementation.

3.2 Key customer benefits

Thus far this case study has focused on the why and how; but what benefits has CellStream given to customers such as NETCOM?

Some of the key benefits gained can be summarised as follows.

- Manage uncertainty

CellStream is now in a position to provide NETCOM's future transatlantic growth and provide the company with unprecedented flexibility and control over its network bandwidth requirements.

- Bandwidth options

PVCs to different destinations can be run on the same cloud which, in turn, means customers can be given high bandwidth Internet access plus other networking services — not just simple Internet. Also the high bandwidth granularity allows customers to have exactly the bandwidth they require and no more.

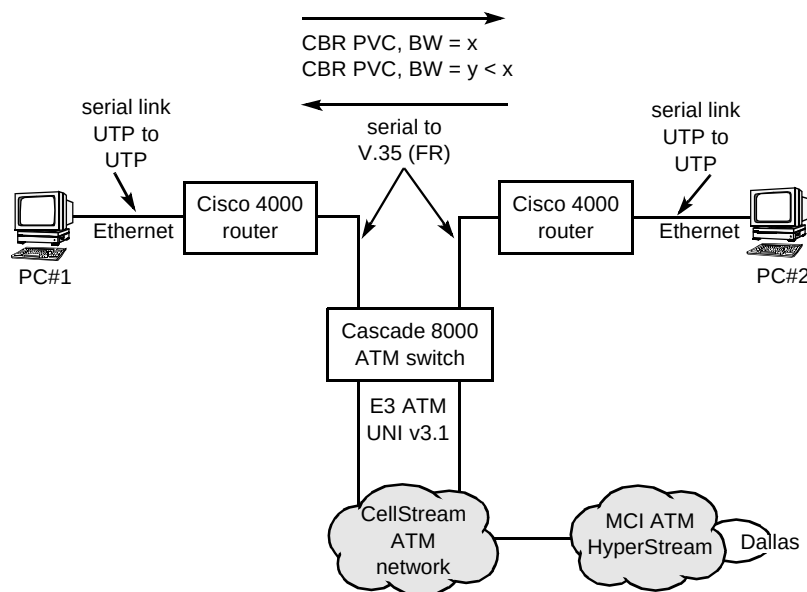


Fig 7 NETCOM test configuration.

- Ability to provide asymmetric circuits

Industry moves fast and there is a need to be able to respond accordingly without having to pay for things that are not needed. This is where the asymmetric bandwidth option is really efficient. It means that the cost savings made can be invested in other parts of the network.

- Cost savings

Cost savings are achieved on the equivalent level of bandwidth on an IPLC solution. As NETCOM only has to pay for its required asymmetric traffic profile and actual bandwidth, it is easy to see that it will save money on the UK-to-US traffic where it only requires maybe half the bandwidth going out to the US compared with that coming in. Using an IPLC it would have to pay for the bandwidth in both directions.

Also, using IPLCs would incur extra costs for inverse multiplexer equipment at both ends.

- Rapid circuit provision

The provision of new VCs in a few days is vital for the efficient management of rapidly growing bandwidth demands such as those required for the Internet. The traditional approach using international private circuits requires customers to accurately forecast the timing of their future needs because typical provision times for new circuits span several months — with CellStream, PVC changes can be done in 8 hours.

4. Conclusions

This paper has outlined BT's first native ATM service, CellStream, and illustrated how customers have been supported in their introduction to the service.

A key lesson learnt thus far is that customers, vendors and BT need to work closely together in implementing technically innovative and cost-effective solutions.

The support provided to NETCOM by BTL and the building of a close relationship resulted in a high level of confidence in BT's technical capability and its commitment to the CellStream service.

Acknowledgements

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References

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Ian Gallagher joined BT as a sponsored student and graduated from Sheffield University in 1973. After working on the development of the System X processor and operating system, in 1985 he established a group investigating future network technologies. Since then he has managed several teams developing BT's first ATM demonstration based on 'Orwell', developing the European ATM pilot network and now developing BT's ATM platform.